

Rate and Conditional Content for Gaussian Vector Sources with Encoder-Observed Context

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Abstract

A stored description has a rate, paid by the decoder, and a conditional content, the least rate at which a party holding side information can re-encode it. For a scalar described variable a companion paper solved the problem in closed form. This paper treats a described vector. An encoder observes a jointly Gaussian source made of a described vector and a correlated context, a decoder reconstructs the vector from the description alone under a total squared-error budget, and a third party holds a noisy copy of the context. We prove the coding theorem for this source: the region of jointly achievable rate and conditional content is the single-letter region, and Gaussian descriptions, indexed by their error covariance, exhaust it. When the coordinates carry independent contexts, the region is a union of sums of scalar regions, and the content-optimal split of the distortion equalizes slopes that we compute in closed form. That split is not reverse water-filling, and the rate-optimal and content-optimal descriptions can describe disjoint sets of coordinates. When the coordinates are coupled, the minimal conditional content is a determinant maximization in the error covariance left to the side-information holder. In a one-dimensional regime, certified by an explicit test, it has a closed form in the largest generalized eigenvalue of a matrix pencil in which the distortion budget enters, and the direction the description reads turns with the budget unless it reads a principal axis of the described vector that is uncorrelated with the context. Principal components, reverse water-filling, and the Gaussian information bottleneck all read fixed directions.

Index Terms

Conditional mutual information, Gaussian vector sources, rate-distortion theory, side information, water-filling.

I. INTRODUCTION

A description of data stored for one party is often kept by another. The decoder that reconstructs from the description pays its rate. A party that keeps the description while holding side information pays its conditional entropy given that side information, the least rate at which that party can re-encode it. The companion paper [1] studies this pair of costs when an encoder observes a jointly Gaussian pair, a scalar described variable Y and a correlated context V , and a third party holds a noisy copy S of the context. It proves the coding theorems, evaluates the minimal conditional content in closed form, characterizes the region of jointly achievable rate and conditional content, and shows that the two costs are minimized by different descriptions.

The present paper takes the described variable to be a vector. A record in a store is rarely one number, and the question becomes which parts of it a description should resolve. The classical answer for the rate, reverse water-filling [2], describes the principal components of the described vector in decreasing order of variance, along fixed directions. We show that the answer for the conditional content is different in kind, and we organize the results around three questions.

The first is whether the single-letter region still has operational meaning. Theorem 3 proves the coding theorem for a vector described variable, extending the Gaussian operational theorem of [1], and Proposition 2 shows that Gaussian descriptions exhaust the region and that each is summarized by its error covariance. The holder's side information enters that summary as added precision.

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The second is how a content-optimal description splits a distortion budget across coordinates that carry independent contexts. Lemma 4 computes the slope and curvature of the scalar function of [1] in closed form and proves it strictly convex, and Theorem 5 shows that the content-optimal split equalizes these slopes. It begins describing a coordinate in decreasing order of $\sigma_j^2(1 + \rho_j^2/\tau_j^2)$ rather than of the variance σ_j^2 , so a small coordinate whose context the holder reads well is described first, and the rate-optimal and content-optimal descriptions can describe disjoint sets of coordinates (Corollary 6).

The third is what happens when the coordinates are coupled, so that no change of coordinates separates them. Theorem 7 shows that the minimal conditional content is the value of a determinant maximization under linear matrix inequalities in the error covariance that remains for the holder, with a unique optimum. Theorem 8 gives the optimum in closed form whenever the optimal description has one dimension, with an explicit certificate for that regime, in terms of the largest generalized eigenvalue of a matrix pencil that holds the source covariance, the conditional covariance left after the side information, and the distortion budget. Its eigenvector is the direction the description reads, and Theorem 9 shows that this direction turns as the budget changes unless it is a principal axis of the described vector that the context does not touch. For a scalar described variable the same pencil returns the closed form of [1].

Related work

Independent Gaussian components and reverse water-filling are classical [2], and Section III uses the same decomposition; what differs is the allocation it produces. Parameterizing a Gaussian description by its error covariance is a standard device in vector Gaussian source coding [3], and Section IV applies it, with the determinant maximization of [4] as the resulting program. The Gaussian information bottleneck [5] projects onto fixed eigenvectors of a normalized conditional covariance, with its tradeoff parameter setting only how many and at what scale; the content-optimal description here reads directions that move with the budget. The single-letter functional is Steinberg's common-reconstruction functional [6] on the source augmented by the context, as noted for finite alphabets by Lapidoth, Malär, and Wigger [7]. The closest Gaussian counterpart, Lu and Xu's vector Gaussian Heegard–Berger problem with common reconstructions [8], constrains the error covariance of the whole source, context included, under a hypothesis that excludes an unpenalized context; [1, Appendix C] shows that its closed form does not continue to that limit even for scalar Y .

The paper is organized as follows. Section II sets up the model and proves the coding theorem. Section III treats independent coordinates and Section IV coupled ones. Section V lists what remains open, and Appendix A describes the numerical checks.

II. MODEL AND THE CODING THEOREM

All logarithms are base two unless written $\ln$. The encoder observes i.i.d. copies of $T = (Y, V)$, jointly Gaussian and centered, with $Y \in \mathbb{R}^m$, $V \in \mathbb{R}^r$, $p = m+r$, and $\Sigma_T \succ 0$. The third party holds $S = V+U$ with $U \sim \mathcal{N}(0, \Sigma_U)$, $\Sigma_U \succ 0$, independent of T and of all channel randomness. Write $F = [I_m \ 0]$ and $H = [0 \ I_r]$ for the coordinate selections, so $Y = FT$ and $V = HT$, and

$$J_S := H^\top \Sigma_U^{-1} H, \quad \Sigma_{T|S} = (\Sigma_T^{-1} + J_S)^{-1},$$

the information that S carries about T and the conditional covariance it leaves. A code consists of an encoder f_n of T^n into a finite set $\mathcal{M}_n$ and a decoder $g_n : \mathcal{M}_n \rightarrow (\mathbb{R}^m)^n$. With $M = f_n(T^n)$ and $\hat{\mathbf{Y}}^n = g_n(M)$, its rate, distortion, and conditional content are

$$R_n = \frac{1}{n} \log |\mathcal{M}_n|, \quad D_n = \frac{1}{n} \sum_{i=1}^n \mathbb{E} \|Y_i - \hat{Y}_i\|^2, \quad L_n = \frac{1}{n} H(M | S^n),$$

and a pair (R, L) is achievable at distortion D when a sequence of codes has $\limsup R_n \leq R$, $\limsup L_n \leq L$, and $\limsup D_n \leq D$; the closure of the achievable set is $\mathcal{RW}(D)$. These are the definitions of [1, Section II] with a vector reproduction. Throughout, $0 < D < \text{tr } \Sigma_Y$; at $D \geq \text{tr } \Sigma_Y$ the zero reproduction gives $R = L = 0$. Write $R_{\min}(D)$ and $L(D)$ for the least R and the least L over $\mathcal{RW}(D)$, the minimal rate and the minimal conditional content; by Theorem 3 they are the minima of $I(T; \hat{\mathbf{Y}})$ and $I(T; \hat{\mathbf{Y}} | S)$ over admissible channels, and a channel attaining one of them is called rate-optimal or content-optimal.

A test channel $p(\hat{\mathbf{y}} | t)$ is admissible when $\mathbb{E}\|\mathbf{Y} - \hat{\mathbf{Y}}\|^2 \leq D$, and the Markov chain $\hat{\mathbf{Y}} - T - S$ holds because the description is formed without S . For a channel, let Σ_{e0} and Σ_e be the error covariances of the best linear estimates of T from $\hat{\mathbf{Y}}$ and from $(\hat{\mathbf{Y}}, S)$. We use one result of [1].

Lemma 1 ([1, Lemma 12 and Remark 13]). *Let $\hat{\mathbf{Y}} \in \mathbb{R}^m$ be centered with finite second moments, let A be its regression matrix on T and $N' = \hat{\mathbf{Y}} - AT$ its residual, and suppose $\text{Cov}(N', S) = 0$. Then $I(T; \hat{\mathbf{Y}}) \geq \frac{1}{2} \log(\det \Sigma_T / \det \Sigma_{e0})$ and $I(T; \hat{\mathbf{Y}} | S) \geq \frac{1}{2} \log(\det \Sigma_{T|S} / \det \Sigma_e)$, and the jointly Gaussian channel $AT + N$ with $N \sim \mathcal{N}(0, \text{Cov } N')$ independent of (T, S) has the same second moments and meets both bounds with equality. If $\text{Cov } N'$ is singular, either both coordinates are infinite or a nonzero combination $\eta^\top \hat{\mathbf{Y}}$ vanishes and, after an invertible linear map of $\hat{\mathbf{Y}}$, may be dropped as a coordinate.*

For an admissible channel, N' is uncorrelated with T by construction and with U by independence, hence with $S = V + U$, so the lemma applies.

Proposition 2. *The union of the quadrants above $(I(T; \hat{\mathbf{Y}}), I(T; \hat{\mathbf{Y}} | S))$ over admissible channels equals the union, over symmetric matrices Σ_{e0} with $0 \prec \Sigma_{e0} \preceq \Sigma_T$ and $\text{tr}(F\Sigma_{e0}F^\top) \leq D$, of the quadrants above*

$$(r(\Sigma_{e0}), \ell(\Sigma_{e0})) = \left(\frac{1}{2} \log \frac{\det \Sigma_T}{\det \Sigma_{e0}}, \frac{1}{2} \log \frac{\det \Sigma_{T|S}}{\det \Sigma_e} \right), \quad \Sigma_e^{-1} = \Sigma_{e0}^{-1} + J_S. \quad (1)$$

Proof. Given an admissible channel, centering $\hat{\mathbf{Y}}$ leaves both coordinates unchanged and does not raise the distortion. If $\text{Cov } N'$ is singular with infinite coordinates the channel contributes no quadrant; otherwise Lemma 1, after dropping vanishing combinations until $\Sigma_N \succ 0$, replaces it by the jointly Gaussian channel $\hat{\mathbf{Y}} = AT + N$ with no larger distortion and no larger coordinates. Then $\Sigma_{e0} = \text{Cov}(T | \hat{\mathbf{Y}})$ has $\Sigma_{e0}^{-1} = \Sigma_T^{-1} + A^\top \Sigma_N^{-1} A$, so $0 \prec \Sigma_{e0} \preceq \Sigma_T$ and $I(T; \hat{\mathbf{Y}}) = r(\Sigma_{e0})$. The pair $(\hat{\mathbf{Y}}, S)$ is a linear observation of T with independent noises (N, U) , so $\Sigma_e^{-1} = \Sigma_{e0}^{-1} + J_S$ and $I(T; \hat{\mathbf{Y}} | S) = \ell(\Sigma_{e0})$. The distortion is at least $\mathbb{E}\|\mathbf{Y} - \mathbb{E}[\mathbf{Y} | \hat{\mathbf{Y}}]\|^2 = \text{tr}(F\Sigma_{e0}F^\top)$. Conversely, given Σ_{e0} , factor $\Sigma_{e0}^{-1} - \Sigma_T^{-1} = G^\top G$, let $W = GT + N$ with $N \sim \mathcal{N}(0, I)$ independent of (T, U) , and reproduce $\hat{\mathbf{Y}} = \mathbb{E}[\mathbf{Y} | W]$. Then $\text{Cov}(T | W) = \Sigma_{e0}$, the distortion is $\text{tr}(F\Sigma_{e0}F^\top) \leq D$, $\hat{\mathbf{Y}} - T - S$ holds, and since $\hat{\mathbf{Y}}$ is a function of W , data processing bounds its coordinates by those of W , which are (1). $\square$

Theorem 3. *For the source of this section and every $0 < D < \text{tr } \Sigma_Y$, $\mathcal{RW}(D)$ equals the union of Proposition 2.*

Proof. The proof of the scalar Gaussian operational theorem [1, Theorem 29] carries over. We give each step and what changes.

Converse. For any code, $\log |\mathcal{M}_n| \geq I(T^n; \hat{\mathbf{Y}}^n) \geq \sum_i I(T_i; \hat{\mathbf{Y}}_i)$ and $H(M | S^n) \geq I(T^n; \hat{\mathbf{Y}}^n | S^n) \geq \sum_i I(T_i; \hat{\mathbf{Y}}_i | S_i)$, by the chain rule and the independence of the pairs (T_i, S_i) , exactly as in [1, Theorem 29]; neither step uses differential entropy or the dimension of the letters, and every term is finite, being at most $I(T_i; S_i) + \log |\mathcal{M}_n|$ with $I(T_i; S_i) = \frac{1}{2} \log \det(\Sigma_V + \Sigma_U) / \det \Sigma_U < \infty$. The mixture of the per-letter channels is admissible at distortion D_n , and by the convexity of both coordinates in the channel [1, Lemma 3 and Remark 4], (R_n, L_n) lies in the union at D_n .

Closedness. Let $(R_j, L_j) \rightarrow (R, L)$ lie in the union at levels D_j with $\limsup_j D_j \leq D$, and pick Σ_j feasible at D_j with $r(\Sigma_j) \leq R_j$ and $\ell(\Sigma_j) \leq L_j$. The Σ_j lie in the compact set $0 \preceq \Sigma \preceq \Sigma_T$, and $\det \Sigma_j \geq \det \Sigma_T 2^{-2R_j}$ is bounded away from zero, so a subsequence converges to some $\Sigma^* \succ 0$, feasible at D by continuity of the trace. The functions r and ℓ are continuous on positive definite matrices, so (R, L) lies above $(r(\Sigma^*), \ell(\Sigma^*))$. This gives closedness at fixed distortion and the passage from D_n to D in the converse.

Achievability. Fix an admissible channel with $I(T; \hat{\mathbf{Y}}) < \infty$; then $\mathbb{E}\|\hat{\mathbf{Y}}\|^2 \leq 2 \text{tr } \Sigma_Y + 2D < \infty$. Steps A, B, and C of [1, Theorem 29] apply with these changes. In Step A the side information is quantized coordinatewise, as there. In Step B the source T and the reproduction $\hat{\mathbf{Y}}$ are quantized coordinatewise at level k (we write k for the quantizer index of [1, Theorem 29, Step B]), and q_η, S_η for the side-information quantizer of Step A, to avoid clashes with $m = \dim Y$, so $\hat{\mathbf{Y}}_k = q_k(\hat{\mathbf{Y}})$ takes finitely many values in $\mathbb{R}^m$ and $\hat{\mathbf{Y}}_k \rightarrow \hat{\mathbf{Y}}$ in L^2 coordinate by coordinate; the surrogate distortion is $d_k(t_k, \hat{\mathbf{y}}) = \mathbb{E}[\|\mathbf{Y} - \hat{\mathbf{y}}\|^2 | T_k = t_k] = \text{tr Cov}(\mathbf{Y} | T_k = t_k) + \|\mathbb{E}[\mathbf{Y} | T_k = t_k] - \hat{\mathbf{y}}\|^2$, which is bounded for each k . The identification of the true and surrogate distortions uses memorylessness only. The convergence of the surrogate distortion to $\mathbb{E}\|\mathbf{Y} - \hat{\mathbf{Y}}\|^2$ uses the martingale convergence $\mathbb{E}[\mathbf{Y} | T_k] \rightarrow \mathbf{Y}$ in L^2 , applied coordinatewise. The weak convergence of the surrogate law of $(\hat{\mathbf{Y}}_k, S_\eta)$ is the argument there with test functions bounded on $\mathbb{R}^m \times q_\eta(\mathbb{R}^r)$ and continuous in the first argument, followed by the lower semicontinuity

of mutual information [9]. Step C is unchanged. The corner $(I(T; \hat{\mathbf{Y}}), I(T; \hat{\mathbf{Y}} | S))$ at distortion D is therefore achievable, and with the converse and Proposition 2 the region is as stated. $\square$

III. PARALLEL DESCRIBED VARIABLES

Suppose the coordinates carry their own contexts and side information: $r = m$, and the triples (Y_j, V_j, S_j) , $j = 1, \dots, m$, are independent, with $\text{Var } Y_j = \sigma_j^2 > 0$, $\text{Var } V_j = 1$, $\mathbb{E}[Y_j V_j] = \rho_j \sigma_j$, $|\rho_j| < 1$, and $S_j = V_j + U_j$, $U_j \sim \mathcal{N}(0, \tau_j^2)$, $\tau_j^2 > 0$. Each triple is an instance of the scalar model of [1] up to the scale of Y_j . Write $\ell_j(\delta) = \frac{1}{2} \log g^*(\delta)$ for the scalar function of [1, Theorem 14(a)] at (ρ_j, τ_j^2) , where g^* is the larger root of

$$P(g) = \delta s g^2 - (\delta + s - \rho^2)g + (1 - \rho^2), \quad s = 1 + \tau^2,$$

extended by $\ell_j(\delta) = 0$ for $\delta \geq 1$, and $L_j(D_j) = \ell_j(D_j/\sigma_j^2)$. Write $\mathcal{RW}_j(D_j)$ for the single-letter region of coordinate j at distortion D_j , the region of [1, Theorem 14(b)] at normalized distortion D_j/σ_j^2 when $0 < D_j < \sigma_j^2$, the quadrant $[0, \infty)^2$ when $D_j \geq \sigma_j^2$, and empty when $D_j = 0$.

Lemma 4. Fix $|\rho| < 1$ and $\tau^2 > 0$, and for $0 < \delta < 1$ let $g = g^*(\delta)$, $k = gs - 1$, and

$$\Delta = \sqrt{(\delta + s - \rho^2)^2 - 4\delta s(1 - \rho^2)} > 0.$$

Then $\ell(\delta) = \frac{1}{2} \log g^*(\delta)$ is twice differentiable on $(0, 1)$, with

$$\ell'(\delta) = -\frac{k}{2 \ln 2 \Delta}, \quad \ell''(\delta) = \frac{(1 - \rho^2)k^3 + \rho^2 \tau^2 g^2 (\tau^4 g + \tau^2(2g + 1) + g + 1)}{2 \ln 2 g \Delta^3} > 0. \quad (2)$$

Hence ℓ is strictly convex and strictly decreasing on $(0, 1)$, $\ell'(\delta) \rightarrow -\infty$ as $\delta \rightarrow 0$, and $\ell'(\delta) \rightarrow -\tau^2/(2 \ln 2 (\rho^2 + \tau^2))$ as $\delta \rightarrow 1$.

Proof. The root is simple: $\partial_g P = 2\delta s g - (\delta + s - \rho^2) = \Delta$ at the larger root. By the implicit function theorem g^* is smooth, and differentiating $P(g^*(\delta)) = 0$ with $\partial_\delta P = gk$ gives $g' = -gk/\Delta$ and $\ell' = g'/(2 \ln 2 g) = -k/(2 \ln 2 \Delta)$. Differentiating once more with $k' = sg'$ and $\Delta' = 2sg - 1 + 2\delta s g'$, and eliminating δ through $P(g) = 0$, which reads $\delta g k = (s - \rho^2)g - (1 - \rho^2)$, gives the second formula, an identity of polynomials in (g, s, ρ^2) that was also confirmed by computer algebra (Appendix A). The first term of its numerator is positive and the second nonnegative, so $\ell'' > 0$. As $\delta \rightarrow 0$, $g^* \rightarrow \infty$ while $\Delta \rightarrow s - \rho^2 > 0$, so $\ell' \rightarrow -\infty$. As $\delta \rightarrow 1$, $g^* \rightarrow 1$ since $P(1) = (\delta - 1)\tau^2 \rightarrow 0$, so $k \rightarrow \tau^2$ and $\Delta \rightarrow |s - (1 - \rho^2)| = \rho^2 + \tau^2$. $\square$

At $\rho = 0$ the lemma returns $\ell'(\delta) = -1/(2 \ln 2 \delta)$, the slope of the classical function. Define the *activation level* of coordinate j ,

$$\kappa_j := -L'_j(\sigma_j^2) = \frac{\tau_j^2}{2 \ln 2 \sigma_j^2 (\rho_j^2 + \tau_j^2)}, \quad (3)$$

which for the classical function would be $1/(2 \ln 2 \sigma_j^2)$.

Theorem 5. Under the model of this section:

(a) The region.

$$\mathcal{RW}(D) = \bigcup_{D_1 + \dots + D_m \leq D, D_j \geq 0} (\mathcal{RW}_1(D_1) + \dots + \mathcal{RW}_m(D_m)),$$

a union of Minkowski sums, attained by product channels that describe each coordinate by its own scalar channel.

(b) The content endpoint. The minimal conditional content is $L(D) = \min \sum_j L_j(D_j)$ over $D_j \geq 0$ with $\sum_j D_j \leq D$, and the minimizing allocation is unique. For $\theta > 0$ let $D_j(\theta) = \sigma_j^2$ if $\theta \leq \kappa_j$, and otherwise let $D_j(\theta) \in (0, \sigma_j^2)$ solve $-L'_j(D_j) = \theta$. The optimal allocation is $D_j = D_j(\theta^*)$ with θ^* the unique level at which $\sum_j D_j(\theta^*) = D$, attained by the product of the channels of [1, Theorem 14(a)] for the described coordinates, rescaled by σ_j , and $\hat{\mathbf{Y}}_j = 0$ for the others.

(c) The order of description. As D decreases from $\sum_j \sigma_j^2$, the content-optimal description begins to describe coordinate j when θ^* exceeds κ_j , that is, in decreasing order of $\sigma_j^2(1 + \rho_j^2/\tau_j^2)$, while the rate-optimal description proceeds in decreasing order of σ_j^2 .

Proof. (a) By Theorem 3 it suffices to identify the single-letter union. Fix an admissible channel and let $D_j = \mathbb{E}(Y_j - \hat{Y}_j)^2$. Each induced channel $p(\hat{y}_j | t_j)$ satisfies $\hat{Y}_j - T_j - S_j$, because given T_j the other coordinates and the channel randomness are independent of U_j . The pairs (T_j, S_j) are independent, so the chain rule gives

$$I(T; \hat{Y}) = \sum_j I(T_j; \hat{Y}, T^{j-1}) \geq \sum_j I(T_j; \hat{Y}_j),$$

$$I(T; \hat{Y} | S) = \sum_j [I(T_j; \hat{Y}, S, T^{j-1}) - I(T_j; S_j)] \geq \sum_j I(T_j; \hat{Y}_j | S_j),$$

using $I(T_j; S, T^{j-1}) = I(T_j; S_j) < \infty$. If some $D_j = 0$ the pair is infinite and there is nothing to prove. Otherwise each pair $(I(T_j; \hat{Y}_j), I(T_j; \hat{Y}_j | S_j))$ lies in $\mathcal{RW}_j(D_j)$, and the regions are upward closed. Conversely, a product of coordinate channels attains the sum of their pairs with equality, by independence, and each $\mathcal{RW}_j(D_j)$ is the union of quadrants above its single-letter pairs [1, Theorem 14(b)].

(b) By (a) and [1, Theorem 14(a)], $L(D) = \min \sum_j L_j(D_j)$. Each L_j is convex and nonincreasing on $[0, \infty)$, strictly convex on $(0, \sigma_j^2)$ by Lemma 4, with left derivative $-\kappa_j$ at σ_j^2 and zero beyond. For a separable convex program with one linear constraint, an allocation is optimal if and only if some $\theta \geq 0$ lies in $-\partial L_j(D_j)$ for every j , with $\theta(\sum_j D_j - D) = 0$. Since $D < \sum_j \sigma_j^2$, $\theta = 0$ is impossible, so $\theta > 0$, the constraint is active, and $D_j = D_j(\theta)$; no D_j is 0 because $-L'_j \rightarrow \infty$ there. Each $D_j(\theta)$ is continuous and nonincreasing, and strictly decreasing for $\theta > \kappa_j$, so $\sum_j D_j(\theta)$ falls continuously and strictly from $\sum_j \sigma_j^2$ at $\theta = \min_j \kappa_j$ to 0 as $\theta \rightarrow \infty$, and θ^* is unique.

(c) $D_j(\theta) < \sigma_j^2$ exactly when $\theta > \kappa_j$, and θ^* increases as D decreases, so coordinates enter in increasing order of κ_j . For the rate the same argument with $\frac{1}{2} \log^+(\sigma_j^2/D_j)$ in place of L_j gives levels $1/(2 \ln 2 \sigma_j^2)$ and reverse water-filling. $\square$

Corollary 6. *Let $m = 2$ with $\sigma_1^2 > \sigma_2^2$ and $\sigma_1^2(1 + \rho_1^2/\tau_1^2) < \sigma_2^2(1 + \rho_2^2/\tau_2^2)$. For every D in the nonempty interval $[\max\{2\sigma_2^2, \sigma_1^2 + D_2(\kappa_1)\}, \sigma_1^2 + \sigma_2^2)$ the rate-optimal description describes coordinate 1 alone and the content-optimal description describes coordinate 2 alone.*

Proof. Reverse water-filling leaves coordinate 2 undescribed and describes coordinate 1 exactly when $\sigma_2^2 \leq \lambda < \sigma_1^2$, that is, for $2\sigma_2^2 \leq D < \sigma_1^2 + \sigma_2^2$. The hypothesis is $\kappa_2 < \kappa_1$, and Theorem 5(b) describes coordinate 2 alone exactly when $\kappa_2 < \theta^* \leq \kappa_1$, that is, for $\sigma_1^2 + D_2(\kappa_1) \leq D < \sigma_1^2 + \sigma_2^2$. Both intervals end at $\sigma_1^2 + \sigma_2^2$ and begin strictly below it. $\square$

An instance makes the size concrete. Take $(\sigma_1^2, \rho_1^2, \tau_1^2) = (1, 0, 1)$, a coordinate whose context is useless, and $(\sigma_2^2, \rho_2^2, \tau_2^2) = (0.6, 0.9, 0.05)$, a smaller coordinate with an informative context read in little noise. Then $\kappa_1 = 0.7213$ and $\kappa_2 = 0.0633$, and the two descriptions describe disjoint coordinates for every $D \in [1.2002, 1.6)$. At $D = 1.4$ the rate-optimal description pays $R = L = 0.1610$ bits, all on coordinate 1, while the content-optimal description pays $L = 0.0197$ bits and $R = 0.3272$ bits, all on coordinate 2: the first pays about eight times the minimal content, the second about twice the minimal rate.

The theorem applies whenever an orthogonal change of coordinates of Y and invertible linear changes of coordinates of V and of S reduce the source to independent triples of this form, with the transformed side information again of the form $S_j = V_j + U_j$ with independent U_j , since squared error is invariant under an orthogonal map applied to Y and $\hat{Y}$ together and both information coordinates are invariant under invertible maps of each argument.

IV. COUPLED DESCRIBED VARIABLES

Return to the general model of Section II. By Proposition 2 the conditional content depends on a description only through Σ_e , the uncertainty about T that remains for the party holding S after it reads the description, and in that coordinate the content endpoint is a concave maximization. Write $h(\Sigma_e) = (\Sigma_e^{-1} - J_S)^{-1}$, which is Σ_{e0} .

Theorem 7. *For $0 < D < \text{tr } \Sigma_Y$,*

$$L(D) = \frac{1}{2} \log \det \Sigma_{T|S} - \frac{1}{2} \max \{ \log \det \Sigma_e : 0 < \Sigma_e \preceq \Sigma_{T|S}, \text{tr}(Fh(\Sigma_e)F^T) \leq D \}, \quad (4)$$

and h is matrix-convex on $\{0 \prec \Sigma_e \preceq \Sigma_{T|S}\}$. For a symmetric Z , $Fh(\Sigma_e)F^\top \preceq Z$ if and only if

$$\begin{bmatrix} Z - F\Sigma_e F^\top & F\Sigma_e J_S^{1/2} \\ J_S^{1/2} \Sigma_e F^\top & I - J_S^{1/2} \Sigma_e J_S^{1/2} \end{bmatrix} \succeq 0, \quad (5)$$

so (4) is a determinant maximization under linear matrix inequalities [4]. The maximizer Σ_e^* is unique and the distortion constraint is active there. A feasible Σ_e is the maximizer if and only if $\text{tr}(Fh(\Sigma_e)F^\top) = D$ and there are $\varpi \geq 0$ and $\Theta \succeq 0$ with

$$\frac{1}{2}\Sigma_e - \varpi \Sigma_{e0} F^\top F \Sigma_{e0} = \Sigma_e \Theta \Sigma_e, \quad \Theta(\Sigma_{T|S} - \Sigma_e) = 0, \quad \Sigma_{e0} = h(\Sigma_e), \quad (6)$$

and then $\varpi > 0$.

Proof. Change of variables. The map $\Sigma_{e0} \mapsto \Sigma_e$ carries $\{0 \prec \Sigma_{e0} \preceq \Sigma_T\}$ onto $\{0 \prec \Sigma_e \preceq \Sigma_{T|S}\}$: $\Sigma_{e0} \preceq \Sigma_T$ if and only if $\Sigma_{e0}^{-1} + J_S \succeq \Sigma_T^{-1} + J_S$, and conversely $\Sigma_e \preceq \Sigma_{T|S}$ gives $\Sigma_e^{-1} - J_S \succeq \Sigma_T^{-1} \succ 0$. With Proposition 2 and Theorem 3 this gives (4).

The matrix inequality and convexity. The matrix $J_S^{1/2} \Sigma_{T|S} J_S^{1/2}$ has the same nonzero eigenvalues as

$$\Sigma_{T|S}^{1/2} J_S \Sigma_{T|S}^{1/2} \prec \Sigma_{T|S}^{1/2} (\Sigma_T^{-1} + J_S) \Sigma_{T|S}^{1/2} = I,$$

so on the feasible set $J_S^{1/2} \Sigma_e J_S^{1/2} \prec I$, and the Woodbury identity gives

$$h(\Sigma_e) = \Sigma_e + \Sigma_e J_S^{1/2} (I - J_S^{1/2} \Sigma_e J_S^{1/2})^{-1} J_S^{1/2} \Sigma_e.$$

By the Schur-complement criterion [10, Sec. A.5.5], $Fh(\Sigma_e)F^\top \preceq Z$ if and only if (5) holds. The same criterion with F replaced by I_p and Z a symmetric $p \times p$ matrix shows that $h(\Sigma_e) \preceq Z$ if and only if a block matrix affine in (Σ_e, Z) is positive semidefinite. Averaging two such instances, with $Z_i = h(\Sigma_{e,i})$, gives $h((1-t)\Sigma_{e,0} + t\Sigma_{e,1}) \preceq (1-t)h(\Sigma_{e,0}) + th(\Sigma_{e,1})$.

Existence, uniqueness, activity. For small $\epsilon > 0$ the description with $\Sigma_{e0} = \epsilon \Sigma_T$ has distortion $\epsilon \text{tr} \Sigma_Y < D$, so the feasible set has an interior point. The objective is bounded above on the bounded set $\Sigma_e \preceq \Sigma_{T|S}$, its superlevel sets are closed and bounded away from singular matrices, and so the maximum is attained; $\log \det$ is strictly concave and the feasible set convex, so the maximizer is unique. If the distortion were slack at the maximizer, moving toward $\Sigma_{T|S}$ would stay feasible and strictly raise $\log \det \Sigma_e$: $\Sigma_{T|S}$, with distortion $\text{tr} \Sigma_Y > D$, is infeasible, so $\Sigma_e \neq \Sigma_{T|S}$ and $\text{tr}(\Sigma_e^{-1}(\Sigma_{T|S} - \Sigma_e)) > 0$.

Optimality conditions. Slater's condition holds, so the Karush–Kuhn–Tucker conditions, stationarity, complementary slackness for both constraints, and nonnegativity of the multipliers, are necessary and sufficient. The differential of h is $dh = h \Sigma_e^{-1} d\Sigma_e \Sigma_e^{-1} h$, so the gradient of $\text{tr}(FhF^\top)$ is $\Sigma_e^{-1} \Sigma_{e0} F^\top F \Sigma_{e0} \Sigma_e^{-1}$, and stationarity of $-\frac{1}{2} \ln \det \Sigma_e + \varpi \text{tr}(FhF^\top) + \text{tr}(\Theta(\Sigma_e - \Sigma_{T|S}))$, multiplied on both sides by Σ_e , is the first condition of (6) (the factor $\ln 2$ is absorbed in ϖ). When the distortion constraint is active, its complementary slackness holds for every $\varpi \geq 0$, which gives the stated test; the maximizer has an active constraint, so the test is also necessary. If $\varpi = 0$, the first condition gives $\Theta = \frac{1}{2} \Sigma_e^{-1} \succ 0$, and the second forces $\Sigma_e = \Sigma_{T|S}$, which is infeasible; so $\varpi > 0$. $\square$

When the optimal description has one dimension, the program has a closed form. Write $\Delta = \text{tr} \Sigma_Y - D$ for the reduction in error the description must buy, and

$$A(\Delta) = \Sigma_T F^\top F \Sigma_T - \Delta \Sigma_T.$$

A one-dimensional description is $W = b^\top T + N$ with $b \neq 0$ and $N \sim \mathcal{N}(0, \sigma^2)$ independent, reproduced by $\mathbb{E}[Y | W]$. Its read direction is b , and the direction in which it reduces the error of the reproduction is $F \Sigma_T b$.

Theorem 8. (a) A one-dimensional description meets the distortion constraint with finite content if and only if $\Delta < \lambda_{\max}(\Sigma_Y)$, and the least content among such descriptions is

$$L_{\text{ld}}(D) = \frac{1}{2} \log \left(1 + \frac{\Delta}{\mu_{\max}(\Delta)} \right), \quad (7)$$

where $\mu_{\max}(\Delta) > 0$ is the largest root of $\det(A(\Delta) - \mu \Sigma_{T|S}) = 0$. It is attained with b a corresponding generalized eigenvector and $\sigma^2 = |F \Sigma_T b|^2 / \Delta - b^\top \Sigma_T b$, at which the distortion constraint is active.

(b) Suppose $\Delta < \lambda_{\max}(\Sigma_Y)$, and let Σ_{e0} and Σ_e be the error covariances of the description of (a), and $\Gamma(\varpi) = \frac{1}{2}\Sigma_e - \varpi \Sigma_{e0} F^\top F \Sigma_{e0}$. Then $L(D) = L_{1d}(D)$ if and only if there is $\varpi \geq 0$ with $\Gamma(\varpi) \succeq 0$ and $\Gamma(\varpi)b = 0$, and then $\varpi = \frac{1}{2}b^\top \Sigma_e b / |F \Sigma_{e0} b|^2$.

(c) For $m = 1$, $L(D) = L_{1d}(D)$ at every $0 < D < \text{Var } Y$ [1, Proposition 24]. In the normalization of [1, Section III], with scalar context, $\text{Var } Y = \text{Var } V = 1$, and $\Sigma_U = \tau^2$, the closed form of [1, Theorem 14(a)] is $g^* = 1 + (1 - D)/\mu_{\max}(1 - D)$.

Proof. (a) For the description W , $\text{Cov}(T | W) = \Sigma_T - \Sigma_T b b^\top \Sigma_T / (b^\top \Sigma_T b + \sigma^2)$, so the distortion of $\mathbb{E}[Y | W]$ is $\text{tr } \Sigma_Y - |F \Sigma_T b|^2 / (b^\top \Sigma_T b + \sigma^2)$, and the constraint reads $\sigma^2 \leq |F \Sigma_T b|^2 / \Delta - b^\top \Sigma_T b$. The content $I(T; W | S) = \frac{1}{2} \log(1 + b^\top \Sigma_{T|S} b / \sigma^2)$ decreases in σ^2 , so at the largest admissible σ^2 , where the constraint is active, it is $\frac{1}{2} \log(1 + \Delta b^\top \Sigma_{T|S} b / b^\top A(\Delta) b)$, finite exactly when $b^\top A(\Delta) b > 0$. Minimizing over b is maximizing the Rayleigh quotient $b^\top A(\Delta) b / b^\top \Sigma_{T|S} b$ of a symmetric pencil with $\Sigma_{T|S} \succ 0$, whose maximum is $\mu_{\max}$. Some b has $b^\top A(\Delta) b > 0$ if and only if $\Sigma_T^{1/2} F^\top F \Sigma_T^{1/2} \not\leq \Delta I$, and the nonzero eigenvalues of that matrix are those of $F \Sigma_T F^\top = \Sigma_Y$.

(b) The description of (a) has an active distortion constraint and $\Sigma_e^{-1} = \Sigma_{T|S}^{-1} + b b^\top / \sigma^2$, so $\Sigma_{T|S} - \Sigma_e$ is a positive multiple of $\Sigma_{T|S} b b^\top \Sigma_{T|S}$ and $\Sigma_e^{-1} \Sigma_{T|S} b$ is a positive multiple of b . With $\Theta = \Sigma_e^{-1} \Gamma(\varpi) \Sigma_e^{-1}$, the conditions (6) at this point therefore read $\Gamma(\varpi) \succeq 0$ and $\Gamma(\varpi)b = 0$, and Theorem 7 makes them necessary and sufficient. Taking $b^\top \Gamma(\varpi) b = 0$ gives ϖ , and $F \Sigma_{e0} b \neq 0$ because $\Gamma(\varpi)b = 0$ with $\Sigma_e \succ 0$.

(c) The first statement is [1, Proposition 24]. For the second, substituting $\mu = (1 - D)/(g - 1)$ into $\det(A(1 - D) - \mu \Sigma_{T|S})$ gives $-(1 - D)(1 - \rho^2)P(g)/(s(g - 1)^2)$, a computation confirmed by computer algebra (Appendix A). The root $g^* > 1$ of P corresponds to the unique positive μ , which is $\mu_{\max}$; the other root of P lies in $(0, 1)$ and corresponds to a negative μ . $\square$

For the rate, the same computation with $J_S = 0$ makes $\Sigma_{T|S} = \Sigma_T$ and the read direction the principal axis of Σ_Y , whatever the budget (the rate-optimal description itself is one-dimensional when $\Delta \leq \lambda_1(\Sigma_Y) - \lambda_2(\Sigma_Y)$). For the content the budget enters the pencil.

Theorem 9. Let $\mu_{\max}(\Delta)$ be simple and positive on an open interval $\mathcal{I} \subseteq (0, \lambda_{\max}(\Sigma_Y))$, with generalized eigenvector $b(\Delta)$. Then $b(\Delta)$ spans the same line for every $\Delta \in \mathcal{I}$ if and only if $b(\Delta_0) = F^\top \beta$ for some $\Delta_0 \in \mathcal{I}$, where β is an eigenvector of Σ_Y with $\text{Cov}(V, \beta^\top Y) = 0$. Otherwise $b(\Delta)$ is constant on no open subinterval of $\mathcal{I}$.

Proof. A simple eigenvalue of a symmetric pencil with positive definite second matrix, and its eigenvector, are analytic in the parameter [11]. Write $A(\Delta) = A_0 - \Delta \Sigma_T$.

Suppose b is constant on an open subinterval. Differentiating $(A_0 - \Delta \Sigma_T)b = \mu_{\max}(\Delta) \Sigma_{T|S} b$ gives $\Sigma_T b = c \Sigma_{T|S} b$ for a scalar c , and $c = b^\top \Sigma_T b / b^\top \Sigma_{T|S} b > 0$. Then $A_0 b = (\Delta + \mu_{\max}/c) \Sigma_T b$, that is, $F^\top F \Sigma_T b = \lambda b$ with $\lambda = \Delta + \mu_{\max}/c > 0$. So $b = F^\top \beta$ with $\beta = F \Sigma_T b / \lambda$, and $\Sigma_Y \beta = F \Sigma_T F^\top \beta = \lambda \beta$. Multiplying $\Sigma_T b = c \Sigma_{T|S} b$ by $\Sigma_{T|S}^{-1} = \Sigma_T^{-1} + J_S$ gives $J_S \Sigma_T b = (c - 1)b$. The left side lies in the range of $H^\top$ and the right side in the range of $F^\top$, so both vanish: $\Sigma_U^{-1} \text{Cov}(V, Y) \beta = 0$, that is, $\text{Cov}(V, \beta^\top Y) = 0$. This proves the last sentence of the theorem and the forward direction.

Conversely, let $b_0 = F^\top \beta$ with $\Sigma_Y \beta = \lambda \beta$ and $\text{Cov}(V, \beta^\top Y) = 0$. Then $\Sigma_T b_0 = \lambda b_0$ and $J_S \Sigma_T b_0 = 0$, so $\Sigma_{T|S} b_0 = \Sigma_T b_0 = \lambda b_0$ and $A(\Delta) b_0 = (\lambda - \Delta) \Sigma_{T|S} b_0$ for every Δ : b_0 is a generalized eigenvector with eigenvalue $\lambda - \Delta$ throughout. Let $\mathcal{E} = \{\Delta \in \mathcal{I} : \lambda - \Delta = \mu_{\max}(\Delta)\}$. It contains Δ_0 and is closed in $\mathcal{I}$ by continuity. It is open: at $\Delta_1 \in \mathcal{E}$ the eigenvalue $\mu_{\max}(\Delta_1)$ is simple, so for Δ near Δ_1 exactly one eigenvalue of the pencil lies near it, namely $\mu_{\max}(\Delta)$, and $\lambda - \Delta$ is an eigenvalue near $\lambda - \Delta_1 = \mu_{\max}(\Delta_1)$. Since $\mathcal{I}$ is connected, $\mathcal{E} = \mathcal{I}$, and by simplicity $b(\Delta)$ spans the line of b_0 on all of $\mathcal{I}$. $\square$

An instance shows the effect. Take $\Sigma_Y = \text{diag}(1, 0.4)$, a scalar context with $\text{Var } V = 1$ and $\text{Cov}(Y, V) = 0.6 u$ for the unit vector u at 50° , and $\Sigma_U = \tau^2 = 0.05$. The certificate of Theorem 8(b) holds for every $D \geq D_*$, with $D_* = 1.08530$ to five decimal places, out of $\text{tr } \Sigma_Y = 1.4$. The rate-optimal description reduces the error along the first axis at every $D \geq 0.8$, in particular throughout the one-dimensional regime. The content-optimal description reduces it along a direction at 47.48° at $D = 1.38$, 46.76° at $D = 1.3$, 45.07° at $D = 1.2$, and 41.15° at $D = 1.1$. The direction $F \Sigma_T b$ is a function of the read direction b , so the read direction turns as well.

The direction is not determined by the law of (Y, S) either, which is the analogue for directions of [1, Corollary 21]. Four sources with this Σ_Y , the context along the same u , and $|\text{Cov}(Y, V)|^2 / (1 + \tau^2) = 0.3$ share the

law of (Y, S) up to scaling of S . At $(|\text{Cov}(Y, V)|^2, \tau^2) = (0.33, 0.1), (0.36, 0.2), (0.39, 0.3), \text{ and } (0.42, 0.4)$, at $D = 1.3$ and inside the one-dimensional regime of each, their content-optimal descriptions reduce the error along $42.83^\circ, 37.61^\circ, 33.75^\circ, \text{ and } 30.84^\circ$, at contents $0.02366, 0.03525, 0.04206, \text{ and } 0.04649$ bits.

Below $D_\star$ the optimal description has two dimensions, and its value is the solution of (4), which interior-point methods compute to any accuracy.

V. DISCUSSION

For a scalar described variable the content-optimal description differs from the rate-optimal one in how much it loads on the context [1]. For a described vector it differs in which parts of the vector it resolves. With independent coordinates it can resolve different coordinates altogether, and with coupled coordinates it reads a direction that moves with the budget, set by a pencil that holds the decoder's distortion, the holder's information, and the budget together. The rate-optimal description, like principal components and the Gaussian information bottleneck, reads fixed directions.

Several problems are open. A closed form for the coupled function below the one-dimensional regime, and for the regime boundary $D_\star$ as a function of the source, is not known. Both coordinates are convex in the error covariance Σ_{e0} : $-\log \det \Sigma_{e0}$ is convex, and $\Sigma_e = (\Sigma_{e0}^{-1} + J_S)^{-1}$ is matrix-concave in Σ_{e0} , so $-\log \det \Sigma_e$ is convex as well. Every weighted combination of rate and content is therefore a convex function of Σ_{e0} under the linear constraint $\text{tr}(F\Sigma_{e0}F^T) \leq D$, and the coupled rate-content frontier is computable by convex programming; a closed-form description of its weighted minimizers is not known. Stationary vector sources, with spectral versions of the function, and the finite-blocklength behavior of the pair (R_n, L_n) are not treated.

APPENDIX A NUMERICAL CHECKS

The identities and instances were checked independently of the derivations, on a remote workstation, by scripts that accompany the manuscript sources. `verify_parallel.py` confirms the slope and curvature identities of Lemma 4 symbolically and against finite differences, matches the equal-slope allocation of Theorem 5 against a brute-force search, checks that no jointly Gaussian channel correlated across coordinates improves on it, and computes the numbers of the parallel instance. `verify_coupled.py` confirms the scalar identity of Theorem 8(c) symbolically, compares the program (4) with a direct search over Gaussian descriptions on random coupled sources, tests the certificate of Theorem 8(b) against the optimizer, and computes the numbers of the coupled instance, including $D_\star$ by bisection on the certificate. The checks guard against algebraic slips; no proof depends on them.

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